

Review

Ecological change in British broadleaved woodland since 1947

JOHN J. HOPKINS* & KEITH J. KIRBY

Natural England, Northminster House, Peterborough PE1 1UA, UK

Over the last 60 years Britain's broadleaved woodlands have undergone a complex pattern of ecological change. The total extent has expanded from *c.* 676 000 ha in 1947 to *c.* 904 000 ha in 2002, but there has also been significant turnover, with losses of ancient woodland and a gain from new planting. Structural change has occurred due, in part, to change in management. In 1947 21% of the broadleaved resource was classed as coppice, 28% as scrub and only 51% of the area as high forest, compared with 97% high forest in 2002. This has been accompanied by changes in ground flora and the regeneration pattern of tree species, which will impact upon the character of the woodlands. Woods have also become more ecologically isolated because of the decline in semi-natural vegetation in the surrounding countryside, although the long-term impacts of this are poorly understood. Other factors driving ecological change are increased pollution, change in grazing pressures, climate change, alien species and game management. Overall woodland specialist species and those of open habitats tend to be doing less well than woodland generalists. Progress has been made in reversing some adverse impacts such as acid deposition, and action is being taken to reduce the impact of others such as over-grazing by deer. However, some drivers, notably climate change, will be more difficult to address. In the long term integrated management of woods as landscape components whilst meeting economic and societal needs will be required.

In the pre-Neolithic period much of Britain may have been wooded, but by 1900 woodland covered only about 5% of the land surface (Rackham 2003, Smout 2003). During the 20th century this increased to *c.* 12% (Forestry Commission 2003a). Much of this expansion was as large-scale conifer plantations, but broadleaved cover has also increased (see Table 1) through both planting and local spread of trees on to downland and other open habitats following reductions in grazing (NCC 1984).

This increase in net broadleaved woodland cover masks gains and losses. For example, between 1980 and 1998 there was a rise of *c.* 5% in 'broadleaved, mixed and yew woodland', through a 12% gain in area of new woodland, offset by a 7% loss of the woodland present in 1990 (Haines-Young *et al.* 2000).

Peterken (1977) and Rackham (1976) highlighted the much greater importance that should be attached to ancient woodland (sites believed to have been wooded continuously since 1600 AD) compared with woods of more recent origin. Between *c.* 1935 and 1985 approximately 7% of such ancient broadleaved woodland was destroyed and about 38% transformed to plantations, largely of conifers (Roberts *et al.* 1992, Spencer & Kirby 1992).

Since 1985 there have been significant shifts in government policies leading to greater protection for existing woodland, particularly ancient woodland, and more encouragement for broadleaved woodland creation, through general forestry policies, the UK Biodiversity Action Plan and the protected area system (Forestry Commission 1985, 2003b, 2005, Thomas *et al.* 1997, English Nature 1998, Kirby 2003a). Although there have been some further losses of ancient woodland to development and

*Corresponding author.

Email: john.hopkins@english-nature.org.uk

Table 1. Changing area (ha) of broadleaved high forest, coppice and scrub (1947–2002).

		Census date			
		1947	1967	1982	2002
Total woodland		1396 097	1742 955	2108 397	2 544 631
Broadleaved woodland	All types	676 034	727 633	749 333	904 245
	High forest	341 640	349 908	564 418	880 722
	Coppice/coppice with standards	141 698	29 594	39 127	23 523
	Broadleaved scrub	192 695	348 138	145 788	Not used

Forestry Census data based on HMSO (1952), Locke (1970, 1987) and Forestry Commission (2003a).

Scrub: inferior crops where more than half the trees were of poor form, poor timber potential, low diameter or composed of unmarketable species. Not used in the 2002 census. Minimum woodland area to which these data refer: 1947, 2 ha; 1967, 0.4 ha; 1982, 0.25 ha; 2002, 2 ha.

agriculture, many of the conifer plantations on ancient woodland sites are now being restored to native broadleaves (Kirby & May 1989, Spencer 2002, Goldberg 2003, Thompson *et al.* 2003, Thompson & Hope 2005).

The increased area of broadleaved habitat should have allowed woodland species to increase, although recent woodland inevitably takes time to build up a characteristic woodland flora and fauna (Peterken & Game 1984, Ferris-Kaan 1995). In practice woodland species have also been affected by changes in the structure and composition of the woods and in the nature of the surrounding landscape. They have also been affected by atmospheric pollution (acidity, eutrophication), climate change, non-native species, grazing and game management. Each of these aspects is considered here, but excluding effects on birds, which are considered in other papers in this issue.

CHANGES IN WOODLAND STRUCTURE AND COMPOSITION

For at least 2000 years woodland in Britain has been managed to greater or lesser extents (Rackham 2003, Smout 2003). Prior to the 20th century much of the ancient woodland, particularly in England, was managed as coppice or coppice with standards. Often the coppice system was abandoned during the latter part of the 19th century, but the First and Second World Wars led to heavy felling to meet the war demands (HMSO 1952, Richards 2003).

Sources of data

Data on the overall changes in the composition and structure of British broadleaved woodland since 1947 have been assembled primarily from the various

forestry censuses (HMSO 1952, Locke 1970, 1987, Forestry Commission 2003a). There are some differences in the minimum size of woodland considered and the categories of woodland recorded in the different censuses, but these are not likely to significantly affect the broad comparisons made in this paper. The census results are supplemented by various long-term monitoring studies. Most of these are of individual sites, such as those at Lady Park Wood (Gloucestershire) or Clairinsh Island (Loch Lomond) (Peterken & Jones 1987, Backmeroff & Peterken 1989), but Kirby *et al.* (2005) provide data from a re-survey of 103 broadleaved woods spread across Britain. These were first recorded using a random sample of 16 200-m² plots in each wood in 1971; between 1999 and 2002 (the '2001' records) these plots were re-surveyed using the same methods. Data are also drawn from structural records from the Repeat Woodland Bird Survey (Amar *et al.* 2006) and the Countryside Survey (1990, 1998), the latter based on stratified random survey of plants and habitats in 1-km squares (Haines-Young *et al.* 2000).

Patterns of change since 1947

In 1947 70% of the broadleaved woodland was in England, particularly in the southeast, 20% in Scotland and 10% in Wales; proportions in 2002 were similar. This review is inevitably therefore dominated by the changes that have affected lowland England. Not all of these changes will be as relevant to the uplands, particularly to northwest Scotland, but the important ecological differences do not necessarily relate to political boundaries: oakwoods in Cumbria have more in common with those in Argyll or Gwynedd (and with those in Killarney) than they do with lime-hornbeam woods of East Anglia.

Table 2. Contrasting expected age distributions for woods managed as coppice and high forest under a regular felling cycle.

Age category	Percentage area in different age categories	
	Coppice with standards	High forest
Open space (< 5 years)	12	4
Young growth (6–30 years)	62	21
Thicket to early mature (31–100 years)	20	58
Mature (> 100 years)	6	17

Proportion of open space (stands less than 5 years old), young growth (stands 6–30 years), thicket to early mature (31–100 years) and mature (> 100 years) for a coppice with standards system run on a 30-year rotation, with 30% of area occupied by standards; and a high forest system run on a 125-year rotation (rotations lengths based on examples in Evans 1984).

Felling of mature broadleaves continued after 1945, until *c.* 1980, as part of the post-war policies to increase home timber production through replacing broadleaved stands with conifers (NCC 1984, Roberts *et al.* 1992, Spencer & Kirby 1992). However in most woods that remained broadleaved, and particularly post-1980, relatively little management has taken place compared with previous centuries. For example, only about 30% of woodlands on private land in England (which makes up 60% of the British broadleaved resource) have a felling licence or are in some form of woodland grant scheme (Slee *et al.* 2006). In 68 of 103 woods visited in 2001, surveyors reported that there was no recent management activity (Kirby *et al.* 2005).

Many of the areas recorded in the forestry censuses as coppice in 1947 and 1967 have now grown up and are classed as high forest (Table 1). Coppice rotations were usually less than 30 years and hence, in worked coppice, many of the stands were either open or young growth (Evans 1984). Such stands are highly valued in biodiversity terms (Buckley 1992, Fuller & Warren 1993). The shift from coppice to a high forest regime, where rotations may be over 100 years (Evans 1984), inevitably reduces the proportion of the stands that are open space and young growth (Table 2) and hence may reduce associated woodland species.

For example, of six butterfly species associated with clearings in woodlands, three have shown marked declines: a 77% decline since 1970–82 in the case of the High Brown Fritillary *Argynnis adippe* (Asher *et al.* 2001). The higher proportion of the early mature growth stages in high forest regimes may benefit some canopy species (Hambler & Speight 1995), but these forest stages tend to have a less distinctive flora and fauna than those at the beginning and end of the growth cycle (Warren & Key 1991). Generalist deadwood species should benefit from

the increases in deadwood reported (Kirby *et al.* 2005, Amar *et al.* 2006). However, specialist saproxylic species tend to be poor colonists and so may not spread to sites where there has not been a continuity of appropriate conditions (Warren & Key 1991).

Intensive coppice regimes in Scotland appear to have been less common (e.g. Samsom 2005) and had already virtually disappeared by 1947. However, in Scotland, a much higher proportion of the woods was classed as scrub (half the total scrub area of Great Britain) – defined as ‘inferior growth unlikely to develop into a utilisable crop of coppice, poles or timber’ (HMSO 1952). Similar processes of growth and canopy expansion are likely to have taken place as with the coppice stands in England.

The shift to high forest (from scrub or coppice) and changes within the structure of high forest stands are supported in trends in the basal area, size distribution of stems, and canopy cover reported in detailed studies. In 103 broadleaved woods surveyed in 1971 and 2001, Kirby *et al.* (2005) found that basal area of woody stems had increased overall; small stems (5–20 cm diameter breast height, dbh) tended to have declined whereas larger size classes had increased. Using different methods, Amar *et al.* (2006) found no increase in mean basal area in their study of 406 woods, but did note an increase in tree height and of subcanopy cover (4–10 m), which would be consistent with a general growth of trees and shrubs in many woods. Similar patterns of increasing stem sizes and/or increasing basal area have also been shown in long-term monitoring studies at individual sites, for example in Wytham Woods (Oxfordshire) (Kirby & Thomas 2000), Monks Wood (Cambridgeshire) (Crampton *et al.* 1998), Dendles Wood (Devon) (Mountford *et al.* 2001) and Clairinsh Island (Backmeroff & Peterken 1989). At other sites, such trends may then be disrupted by major disturbances such as droughts or storm, for

example Lady Park Wood (Peterken & Jones 1987) and The Mens (West Sussex) (Mountford 2004).

Vascular plant richness tends to increase under woodland gaps, whether caused for example by storms (Buckley *et al.* 1994), ride management (Buckley *et al.* 1997), coppice cuts (Barkham 1992) or clear-fells (Kirby 1990), and decline as woody species cover is re-established (Mitchell & Kirby 1989). In their national woodland survey Kirby *et al.* (2005) found a reduction in ground flora species richness associated with increasing basal area of trees and shrubs (basal area often being closely associated with canopy cover). Woodland specialists declined in frequency more often than generalist species while species increasing in abundance tended to be those associated with semi-shade as opposed to open habitats. Haines-Young *et al.* (2000) found similar evidence for increasing shade in the changes in broadleaved woodland flora recorded in the Countryside Survey 2000 for England and Wales.

These national analyses hide local and regional patterns of change in stand structure, for example the contrasting changes in the old and young growth stands in Lady Park Wood, which reflect past management (Peterken & Jones 1987, 1989). Natural events such as storms also impact at local and regional scales. Quine and Gardiner (2002) report the incidence of

approximately one regionally significant storm every 2 years between 1945 and 2000. The severe storm in early 2005 mainly affected woods in northwest England (C. Quine pers. comm.), whereas that of 1987 had little effect outside southeast England (Kirby & Buckley 1994). Ten of the sites surveyed by Kirby *et al.* (2005) were in the track of the 1987 storm. In 2001 these were more likely to show an increase in plant species richness than sites that were outside the storm track, reflecting a tendency for storms to create gaps in otherwise shaded woods.

Changes in tree and shrub composition of broadleaved woods

Since 1947 the woody composition of broadleaved woods has shown some changes (Table 3). Small fluctuations between survey dates and differences in minor species may not mean much because there is a large area that is not differentiated to species in the reports. However, there does appear to be a decline in the relative contributions of Oak *Quercus robur/petraea* and Beech *Fagus sylvatica* (but less change in absolute area terms) and an increase in Ash *Fraxinus excelsior* and Sycamore *Acer pseudoplatanus*. In the 103 woods surveyed by Kirby *et al.* (2005) the frequency of occurrence of the main species showed

Table 3. Changes in the composition of British broadleaved woodland based on Forestry Commission census data (HMSO 1952, Locke 1970, 1987, Forestry Commission 2003a).

	Area (ha) (% of total broadleaved area in parentheses)			
	1947	1967	1982	2002
Oak	210 069 (31.0)	205 182 (28.2)	190 044 (25.5)	20 6154 (22.7)
Ash	44 261 (6.5)	52 348 (7.2)	79 204 (10.6)	119 232 (13.1)
Beech	67 997 (10.0)	68 056 (9.3)	75 006 (10.1)	76 551 (8.5)
Birch	140 893 (20.8)	170 081 (23.4)	131 391 (17.6)	155 355 (17.1)
Sweet Chestnut	22 678 (3.3)	22 834 (3.1)	29 226 (3.9)	10 800 (1.2)
Sycamore	27 623 (4.1)	31 214 (4.3)	54 291 (7.3)	61 357 (6.8)
Alder	9388 (1.4)	17 327 (2.4)	8363 (1.1)	
Hornbeam	5961 (0.9)	1498 (0.2)	3823 (0.5)	
Poplar	1331 (0.2)	7854 (1.1)	13 590 (1.8)	10 418 (1.1)
Lime	719 (0.1)	729 (0.1)		
Elm	9711 (0.1)	8380 (1.1)	9514 (1.3)	3743 (0.4)
Willow	2426 (0.4)	526 (0.1)	4964 (0.7)	
Norway Maple	89 (0.01)			
Cherry	46 (0.01)			
Hazel	60 305 (8.9)	38 421 (5.3)	11 656 (1.6)	
High forest undifferentiated	373 (0.5)	25 141 (3.4)	94 211 (12.6)	237 111 (26.2)
Coppice undifferentiated	50 408 (7.4)	2672 (0.4)	9287 (1.3)	23 526 (2.6)
Scrub undifferentiated	24 055 (3.5)	75 384 (10.4)	30 600 (4.1)	

Table 4. Composition of broadleaved woodland 1971–2001 based on resurvey of 103 woods (Kirby *et al.* 2005).

	Frequency of occurrence in 103 sites (1648 plots)			
	% of plots		% of sites	
	1971	2001	1971	2001
Oak	53	47	92	92
Beech	16	16	56	70
Sycamore	25	27	66	73
Ash	44	44	85	86
Birch	39	32	81	80
Elm	12	9	46	41
Field Maple	9	8	28	31
Rowan	16	16	58	56
Alder	10	10	46	47
Holly	7	16	45	59
Hazel	44	40	87	83
Hawthorn	24	21	79	76
Elder	11	9	49	45

little change (Table 4). This may indicate therefore that the increases in Ash and Sycamore in part reflect the abundance of these latter two species in new woodland rather than increases within existing sites.

Sweet Chestnut *Castanea sativa* and Hazel *Corylus avellana* were the two most common coppice species in 1947 and their decline in the record since then is likely to be in part due to coppice moving into the high forest category and then being classified by the overstorey tree species rather than by the composition of the coppice.

Individual tree species have shown contrasting changes in their size distribution over recent decades both in a national survey (Kirby *et al.* 2005) and in various individual monitoring studies. Oak nationally showed net loss from all size classes smaller than 50 cm dbh and an increase in the larger size classes. Suckering Elm *Ulmus* spp. was hit by Dutch elm disease and declined in most size classes nationally; Wych Elm (*U. glabra*) showed a similar pattern at Lady Park Wood (Peterken & Mountford 1998). Large Beech declined in the national survey and at Lady Park Wood probably because of drought (Peterken & Mountford 1996), although on some sites it is becoming more frequent and its national area has increased slightly (Table 3). In the understorey, Hazel showed a marked decline in the number of stems in the 5–10 cm diameter class nationally and also in Monks Wood (Crampton *et al.* 1998), whereas the shade-tolerant Holly *Ilex aquifolium*

increased nationally and in The Mens and on Clairinsh (Mountford 2004).

CHANGE IN EXTERNAL INFLUENCES ON BROADLEAVED WOODLANDS

Landscape context

Peterken (1992, 1996) and Rackham (1986) stress that prior to 1940 woods were closely linked to the rest of the landscape. Semi-natural habitats comprised a higher proportion of the landscape between the woods. Trees spread out into the surrounding countryside through hedges and areas of wood-pasture. Grassland and heath came into the wood along rides and glades or at the boundaries.

During the 20th century much of the semi-natural habitat around and in between the woods was lost, particularly in the lowlands of Britain (NCC 1984, Peterken & Allison 1989). Even where semi-natural vegetation survives between the woods its quality may have declined (Haines-Young *et al.* 2000). Therefore, despite the increase in woodland area since 1947 changes in the nature of the landscape matrix may have affected the population dynamics of species within woods through an increase in their ecological isolation. Landscape ecologists have sought to model the significance of semi-natural vegetation between woods for the potential dispersal of woodland species (e.g. Latham *et al.* 2004, Watts *et al.* 2005), but as yet we lack quantitative understanding of the impact of historical loss, above the level of individual species (Bailey *in press*). Nonetheless there are proposals for the development of habitat networks (Peterken *et al.* 1995, Moseley *et al.* 2005) to offset the effects of increased isolation of woods.

Changes in nutrient and pollution environment

Increasing macronutrients, nitrogen and phosphorous, have been shown to be affecting terrestrial habitats in the last 50 years (e.g. Preston *et al.* 2002, Smart *et al.* 2003). These are likely to favour increases in competitive, fast-growing plants at the expense of slower-growing woodland species (Grime *et al.* 1988, Gilliam 2006). Sutton *et al.* (2004) estimated that 90% of woodland was likely to receive atmospheric nitrogen deposition in excess of critical loads, i.e. the maximum amount of pollutants that ecosystems can tolerate without being damaged (NEGAP 2001).

High ammonia deposition on to woodlands can occur close to intensive animal housing (Pitcairn *et al.* 1998). Fertilizer and pesticide overspread may also occur on the margins of woodlands adjoining arable and intensive grassland. Most effects have been reported from a zone 10 m wide (Gove *et al.* 2004), although greater penetration of marginal impact has been reported (Willi *et al.* 2005).

Kirby *et al.* (2005) found no overall shift towards more fertile plant assemblages, but the ground flora species that increased in cover between 1971 and 2001 were more likely to be associated with high nutrient conditions. Species typical of more fertile conditions were also more likely to have increased in woods with intensive surrounding land-use. From the Countryside Survey 2000 study, Haines-Young *et al.* (2000) found increases in plants of more fertile conditions between 1990 and 1998 in broadleaved woodland in England and Wales, but not in Scotland. However, this effect may not be solely due to increased nutrient inputs because their study included plots in new woodland created between 1990 and 1998, which had higher fertility scores than the older woodland plots.

Lowland broadleaved woodland vegetation in Britain may be less susceptible to nutrient enrichment than other semi-natural habitats such as heathland, because a greater proportion of woodland species can grow under moderate to high fertility conditions (Smart *et al.* 2003). There may also be some uncoupling of the linkage between nutrient enrichment and floristic change due to the stress caused by the high shade in woodland (Kirby *et al.* 2005).

High levels of acid deposition, particularly sulphur and nitrogen oxides, have been experienced in parts of Britain since the late 19th century (Bates 2002), but acidifying deposition more than halved over large areas of Britain between 1985 and 1999 (NEG-TAP 2001). The onset of high levels of acid deposition pre-dates most monitoring, so that recent studies appear to report recovery from its effects, for example increases in woodland soil pH (Kirby *et al.* 2005) and slight evidence of a decrease in woodland plants of acid soils relative to those of base-rich soils, between 1987 and 2004 (Braithwaite *et al.* 2006). Foliose and fruticose lichen species, which are highly sensitive to acidity, have expanded in parts of Britain where reductions of acid deposition have been most marked (NEG-TAP 2001, Bates 2002). Many moths associated with lichens have increased despite large overall declines in moth abundance (Conrad *et al.* 2004).

Climate change

In the 20th century the British climate has changed, with a c. 1 °C increase in temperature in central England, a lengthening of the growing seasons, wetter winters and slightly drier summers (Hulme *et al.* 2002). The responses of species and communities may be reflected in terms of changes in phenology (i.e. the timing of seasonal events), range, abundance or disturbance regime.

There has been a general advance in spring events including date of first flowering of plants (Fitter & Fitter 2002), first leafing dates of trees (Sparks *et al.* 1997), flight times of moths and butterflies (Sparks & Yates 1997, Woiwod 1997, Roy & Sparks 2000, Burton & Sparks 2002), and first appearance of hoverflies (Morris 2000).

Increasing temperatures are expected to result in northward expansion of range in those species which reach their northern distribution limit in Britain, and this has been reported for birds, butterflies, flowering plants and many other taxonomic groups (Thomas & Lennon 1999, Warren *et al.* 2001, Hickling *et al.* 2005, 2006). Hickling *et al.* (2006) also report a mean upwards altitudinal shift of range in their survey of 16 taxonomic groupings. A poleward range retraction of taxa with their southern limit in Britain has been reported for two northern butterfly species (Franco *et al.* 2006). Amongst woodland species, there has been considerable northward range expansion by the Speckled Wood Butterfly *Pararge aegeria* (Hill *et al.* 1999).

More localized distribution responses may be shown through species changing their habitat preferences. Woodland, through its structure, creates a variety of microclimates, even on flat sites. Butterflies currently using 'hotspots' in glades may shift their behaviour and become more common under canopy shade, as in southern Europe (Thomas 1991).

Kirby *et al.* (2005) tested change in mean cover of 65 abundant ground flora species over the period 1971–2001; 13 species showed reduced cover associated with lengthening growing season and four had increased cover. A larger pool of species was available for analysis of change in frequency. This revealed positive changes in 47 species correlated with January–March temperature change whereas only four showed a negative relationship. Braithwaite *et al.* (2006) found a small decline of northern woodland plant species, which may be linked to climate.

Dry summers are expected to become more frequent (Hulme *et al.* 2002). After the 1976 drought

thousands of Beech died in the New Forest (Tubbs 2001). Peterken and Mountford (1996) showed from their long-term study at Lady Park Wood that droughts can have long-lasting effects on the growth of Beech, with trees still dying 15 years after the event. Even relatively small changes in the frequency of extreme events could therefore have major impacts on some woodland systems. Lesser events might change the competitive balance of major canopy trees such as Beech, Oak and Ash (Broadmeadow & Ray 2005).

The changes in individual plant species, the climatically linked changes in abundance of butterfly species identified by Roy *et al.* (2001) and the increase in southern moth species reported by Conrad *et al.* (2004) may indicate that a general change in the structure of plant and animal communities is starting to occur. This in turn will have implications for conservation policy and practice (Harrison *et al.* 2001, Wesche *et al.* 2006).

Large herbivores in woods

The impact of large herbivores in the pre-Neolithic forests is a matter of current debate (Vera 2000, Hodder *et al.* 2005, Mitchell 2005, Rackham 2006). However, for the last thousand years in Britain the dominant large herbivores have been domestic stock or deer whose populations and distribution have been largely determined by humans (Yalden 1999). Wood-pasture has long been recognized as a valuable form of ancient woodland in the lowlands (Kirby *et al.* 1995), but more recently it has been recognized as having been widespread in the uplands as well (Stiven & Holl 2004). Given the varying history of large herbivores in broadleaved woodland it is not surprising that both too much and too little grazing may have undesirable impacts on the biodiversity of our woods (Williams 2006).

In upland oakwoods past coppicing and grazing by sheep have created distinctive woodland structures typified by high densities of Oak, a rich bryophyte flora, relatively little shrub layer and frequently little sign of regeneration (Ratcliffe 1977, Rodwell 1991, Palmer *et al.* 2004, Baarda 2005). Sheep numbers have tended to increase in the uplands in much of the last 50 years (Fuller & Gough 1999), but in the last decade, at least locally, there have been some reductions as a result of agri-environment schemes. Where grazing is reduced or eliminated, e.g. through fencing to stimulate regeneration, there is often an initial increase in the herb layer, with spread of

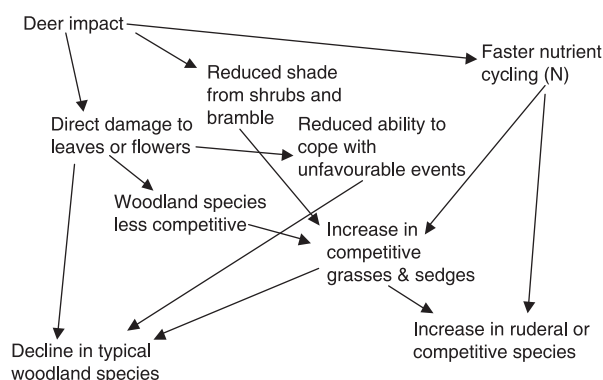


Figure 1. Possible pathways by which deer may have influenced the vegetation changes in Monks Wood, Cambridgeshire.

palatable species such as Bramble *Rubus fruticosus*, as in the long-established enclosure at Wistman's Wood (Devon) and in the woods of Killarney (Kelly 2005). At other sites it may be Wood-Rush *Luzula sylvatica* or dense tall grass mats which develop. Holly or Rowan *Sorbus aucuparia* may spread to create dense understoreys. A denser field layer may reduce the characteristic moss carpet, while the expanding shrub layer may then shade out many of the field layer species (Mitchell & Kirby 1990, Hodgetts 1993, Kirby 2001).

In many lowland broadleaved woods a significant change in the last 50 years has been the spread of deer (White *et al.* 2004, Ward 2005). Coppicing has become increasingly difficult to sustain, even for purely conservation reasons, because stool regrowth is poor without fencing or control of deer densities (Kirby 2003b). The shrub layer tends to be reduced and palatable ground flora species such as Bramble decline in abundance (Kirby 2004). Overall plant species richness may or may not be affected, because woodland species may be replaced by increases in ruderal or grassland species (Kirby 2001) (Fig. 1). Increasing deer populations have also been implicated in changes in populations of small mammals, invertebrates and birds (see papers in Fuller & Gill 2001, Cooke 2006).

Non-native species

Amongst semi-natural habitats woodland has the largest number of recorded non-native species in England (Hill *et al.* 2005), and, after grasslands, is the second most impacted semi-natural habitat in Scotland (Welch *et al.* 2001). Eight of 21 alien species

identified by Hill *et al.* (2005) as having a major negative effect are primarily or exclusively found in woodland.

Non-native species interact with native taxa in a complex way, for example through:

- structural change to habitat, e.g. invasion by *Rhododendron ponticum* (Rotherham & Read 1988);
- herbivory, e.g. Muntjac *Muntiacus reevesi* and Sika Deer *Cervus nippon* (Fuller & Gill 2001);
- disease, e.g. Dutch elm disease *Ophiostoma ulmi* and *O. novo-ulmi* (Brasier & Buck 2001);
- predation, e.g. possible nest predation of Hawfinch *Coccothraustes coccothraustes* by Grey Squirrel *Sciurus carolinensis* (Amar *et al.* 2006);
- competition, e.g. Grey Squirrel and Red Squirrel *Sciurus vulgaris* (Gurnell *et al.* 2004);
- disease vectors, e.g. Grey Squirrel are vectors of squirrel pox to which native Red Squirrel are susceptible (Tompkins *et al.* 2002);
- hybridization and introgression, e.g. Sika Deer and Red Deer *Cervus elaphus* (Goodman *et al.* 1999).

The accidental spread of alien species needs to be seen in perspective, given that some have become major problems but most have not. In the British countryside non-native plants are mainly found in disturbed fertile habitats (Maskell *et al.* 2006). Non-native species may locally support highly valued species; for example, orchards support the Noble Chafer *Gnorimus nobilis* (Whitehead 2003) and old Sweet Chestnut may host rare saproxylic invertebrates (Buckley & Howell 2004). Historically greater nature conservation losses have occurred through the deliberate encouragement of non-native species such as Sitka Spruce *Picea sitchensis* (NCC 1984, 1986). As a result of climate change some species may now be spreading naturally into areas where they have traditionally been considered non-native, for example Beech in northern England (Wesche *et al.* 2006).

Game management

During the 19th century, when management for wood and timber was in decline, the increased interest in managing woods for sport (fox hunting and pheasant shooting) led to woods being retained on many lowland estates. However, there was also widespread persecution that resulted in the reduction and in some cases elimination of mammals and birds considered to be detrimental to sporting interests (Yalden 1999). In woods managed for Pheasant *Phasianus colchicus* shooting (Gray 1986, Robertson

1992) there has often been manipulation of the tree and shrub layers to increase cover at strategic points; planting of non-native shrubs for cover; and erection of pens in which the birds are kept, often at high densities, prior to release. One in 12 woods in England may contain a Pheasant release pen. In heavily stocked pens the ground flora is altered as a consequence of the disturbance and increased fertility resulting from the presence of the birds (Sage *et al.* 2005). The net effect of game management on biodiversity in woodland has, however, probably been generally positive (Bealey & Robertson 1992, Robertson 1992).

DISCUSSION

Since 1947 Britain's woodlands have undergone ecological change of a complexity and character that is historically unparalleled. Many of these changes (increases in wooded area, growth within forests, increased herbivory by deer, effects of pollution and climate change) are common across Europe (Marell & Leitgeb 2005, FAO 2006). In Britain these changes have been largely driven by:

- (1) government policy, particularly where supported by financial incentives;
- (2) changing markets for forest products and services; and
- (3) the environmental impacts of sectors other than forestry.

Considered at a national level, some of the ecological drivers have had a continuous or increasing ecological effect in the past 60 years. These include, changing shade due to canopy closure, nutrient enrichment, non-native species invasion, deer grazing and climate change as well as the expansion of total broadleaved woodland cover.

However, in recent years other ecological drivers have shown a decline; for example, conversion of broadleaved woodland to other land-use increased in the decades after 1947, but has declined since the mid-1980s, and acid deposition has also declined since the 1980s. These reductions in adverse impacts can be seen as successes for biodiversity conservation, although the effects of earlier damage are by no means fully reversed.

The individual drivers and their responses also differ in intensity across Britain, within woods and even within stands. At the national scale, upland woods may still be more heavily grazed than lowland ones, but recent increases in grazing have been most noticeable in lowland woods (Kirby *et al.* 2005). The

shift away from coppice systems was more significant for southeast England than for northwest Scotland. At a smaller scale, in mature stands the shade from shrub and tree layers is an important determinant of the ground flora, but in the open phase following felling or storm damage, competition within the ground flora becomes more significant (Kirby 1990).

Current woodland biodiversity is contingent upon historical events, past climates and land-use patterns. Old and new woods differ, hence the value placed by conservationists on ancient woodland (Peterken 1977). However, there may also be differences between new woodland created since 1947 and earlier plantations, as new woodland will usually be developing on a richer agricultural soil and in a different climate (Haines-Young *et al.* 2000, Hulme *et al.* 2002).

The interactions between the various drivers and the effects of contingency make it impossible to predict the precise outcome of management to improve woodland biodiversity. Despite this, removing obvious adverse impacts on biodiversity where possible should be a high priority, for example through locally managing deer populations, and at national level through improved controls on alien species introductions. This should increase the resilience of woodlands in the face of those drivers, such as climate change, where mitigation options are limited. For these drivers, changes to woodland and landscape management will be required if we are to maintain and develop our woodland biodiversity, as proposed, for example, through the various woodland network proposals (Latham *et al.* 2004, Moseley *et al.* 2005).

In turn this will require re-connecting our woods with the changed socio-economic landscape by finding new markets for woodland products or ways of translating their values for biodiversity, landscape or recreation into returns to the individual owner (Slee *et al.* 2006). Woodland policies have changed several times during the last 60 years (Kirby 2003b, Richards 2003) and further shifts seem likely before trees that started to grow in 1947 reach maturity.

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Received 17 July 2006; revision accepted 26 February 2007.